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NATIONAL UNIVERSITY
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P R O J E C T P R O P O S A L

Guardian Knot:

A C++ Based Automated Persistence Safety System for Emergency Monitoring



Subject : Computer Programming

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G R O U P M E M B E R

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Abstract:

Digital safety and emergency response are critical in modern software development. While many apps exist for active tracking, there is a gap in "passive monitoring" where a user might be unable to call for help due to an accident or restriction. Guardian Knot caters to this by using a "Dead Man's Switch" logic implemented in C++. It monitors user activity via periodic check-ins; if a user fails to verify their safety within a set timeframe, the system automatically triggers an emergency protocol. We expect to achieve a reliable, low-resource safety tool for hostelites and solo travelers using core C++ programming.

Introduction:

The problem addressed is "silent emergencies" situations where a person is incapacitated and cannot reach their phone. Globally, solutions like wearable panic buttons or SOS apps exist, but they require the user to take an action during the crisis. What remains unsolved is a system that acts when the user cannot. This project implements a time-threshold logic where the system assumes a "danger state" if no "safe signal" is received. Using C++ File Handling, the program ensures that even if the computer restarts, the safety timer remains accurate and persistent.

Problem Statement:

To develop a C++ persistence-based monitoring system that triggers emergency alerts automatically when user-defined safety check-in intervals are exceeded.

Hypothesis:

Our solution uses a console-based interface where users register their details and emergency contacts, which are stored in .txt files using File Handling. The system uses the <ctime> library to calculate the difference between the "Last Seen" timestamp and the "Current Time".

Technical Process:

The program runs a loop that compares Current Time – Saved Time. If the result > 172,800 seconds (48 hours), it triggers a red-alert screen and generates an emergency log file.

Difficulties:

A major challenge is ensuring the program handles incorrect password attempts without locking the actual user. We will cater to this by implementing a "Duress Pin" and using Input Validation in our logic.

Workflow Algorithm:

- **Start:**

The program initializes and checks for the existence of user_config.txt using the ifstream class.

- **Registration:**

If the configuration file is not found, the registration() function is called to collect and store the user's name, emergency contact, and secret passwords.

- **Time Retrieval:**

If the file exists, the system reads the "Last Seen" timestamp and fetches the current system time using the <ctime> library.

- **Logical Comparison:**

The program calculates the time difference (Gap = Current Time - Saved Time).

Security State:

1. Safe State:

If the gap is less than 24 hours, the system displays a "Safe" status and prompts for a password to reset the timer.

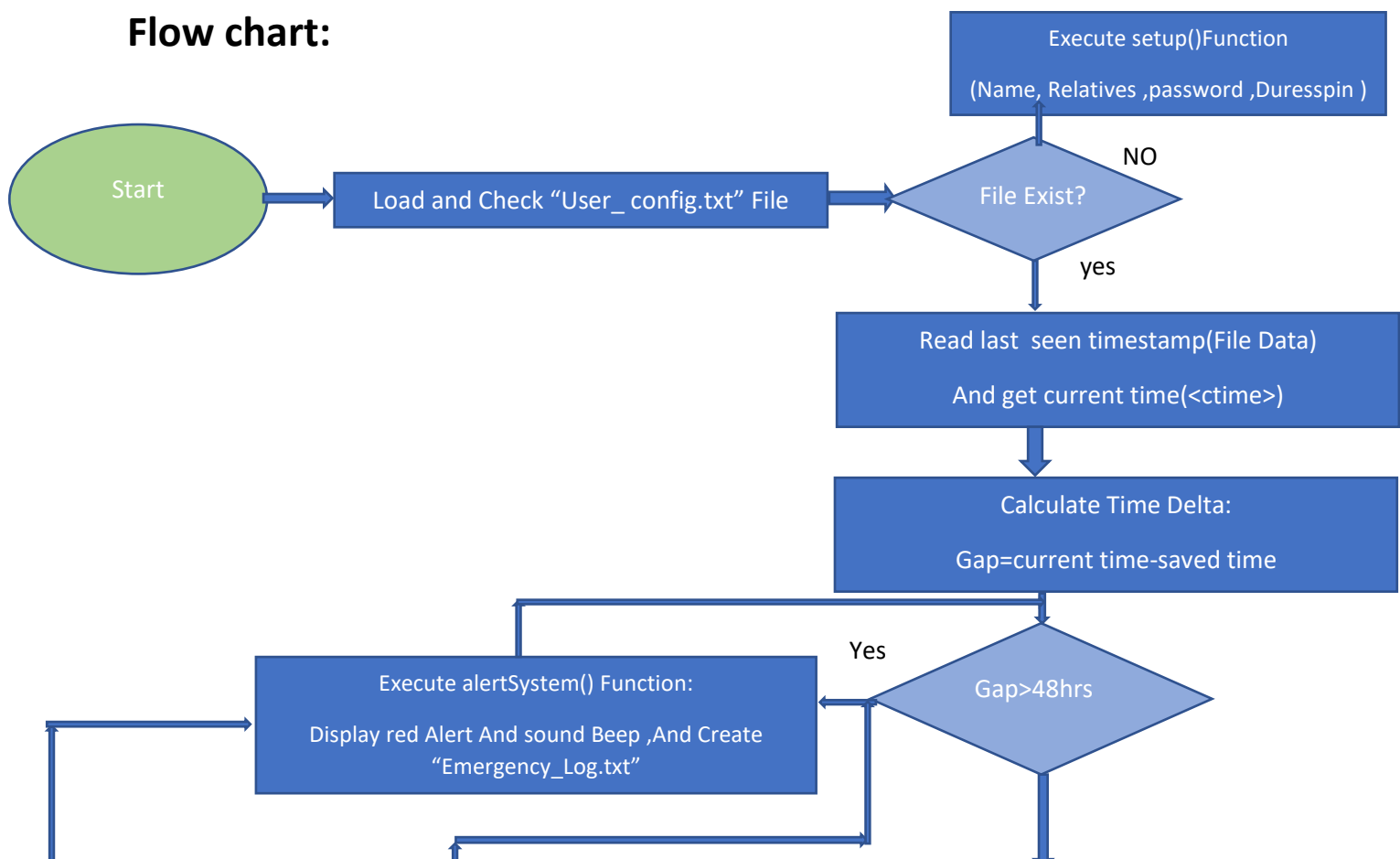
2. Danger State:

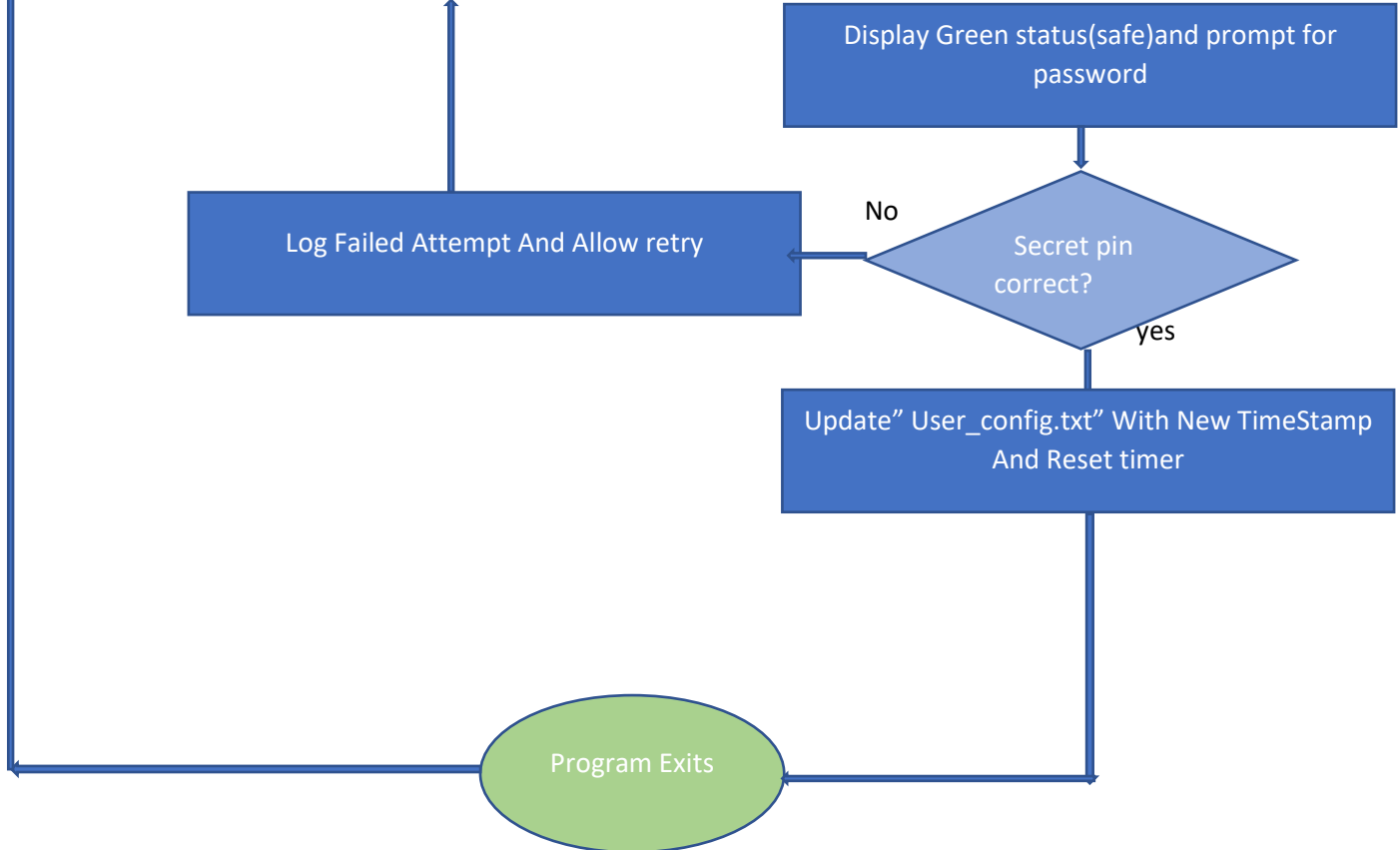
If the gap exceeds 48 hours, the alertSystem() function is triggered, displaying a warning and generating an EMERGENCY_LOG.txt file.

Termination:

The system updates the file with the new timestamp upon successful verification and exits.

Flow chart:





Conclusion:

The "Guardian Knot" project addresses the gap in passive emergency monitoring for individuals in isolation. By utilizing core C++ concepts such as Functions, Arrays, and File Handling, we have developed a system that operates autonomously to ensure user safety. This project fulfills the requirements of the course by demonstrating effective application-level programming and reliable data persistence.

References:

- C++ Project Guidelines, BEE-17D Course Material, NUST.
 - Project Proposal Template D, Computer Programming Department.
 - Standard C++ Library Documentation for fstream and ctime.
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